Chapter 1
Matrices and Systems of Equations

1.5 Elementary Matrices

Objectives

- Elementary Matrices
- Triangular Matrices

Elementary matrices

- In this section, we view the process of solving a linear system in terms of matrix multiplications rather than row operations. [A|b] → → [|]
- Given an $m \times n$ linear system $A\mathbf{x} = b$, we can obtain an equivalent system by multiplying both sides of the equation a nonsingular $m \times m$ matrix:

$$MA\mathbf{x} = Mb$$

- Can transform the system $A\mathbf{x} = b$ to a simpler form that is easier to solve by multiplying a sequence of nonsingular matrices to the equation.

Elementary Matrices

Definition

An **elementary matrix** is one that can be obtained from the identity matrix I_n through a single elementary row operation.

Example 1

Three elementary row operations:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

R2 ↔ R3 $\rightarrow$ $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

5R2 $\rightarrow$ $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

R2+ 2R1 $\rightarrow$ $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Eg. Is $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$ an elementary matrix?

$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{3R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 + 2R_1} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ✗

→ only one operation

Type I: Interchange

Type II: Scaling

Type III: Replacement

Elementary Matrices

- Type 1. A matrix obtained by interchanging two rows of I.

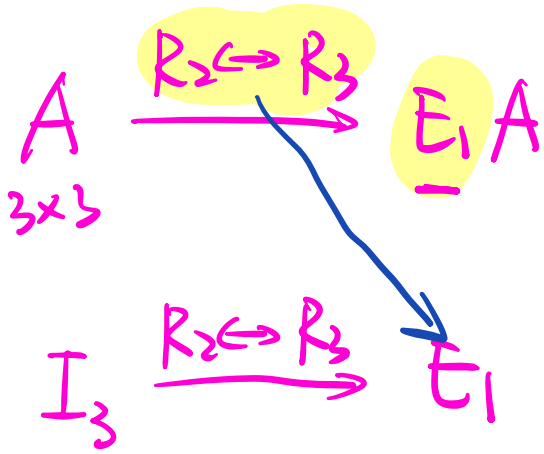
E.g.,

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$AE_1 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{13} & a_{12} \\ a_{21} & a_{23} & a_{22} \\ a_{31} & a_{33} & a_{32} \end{bmatrix}$$

Multiplying on the left by E1 interchanges rows R2 and R3 of A, and multiplication by E1 on the right interchanges columns C2 and C3 of A.



row operation.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{C_2 \leftrightarrow C_3} E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$A \xrightarrow{R_2 \leftrightarrow R_3} E_1 A$$

$$A \xrightarrow{C_2 \leftrightarrow C_3} A E_1$$

Eg. $A_{3\times4} \xrightarrow{R_1 \leftrightarrow R_3} E_1 A_{3\times4} \xrightarrow{C_2 \leftrightarrow C_4} (E_1 A)_{3\times4} E_2$

E_1 : 3×3 ; E_2 : 4×4

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{C_2 \leftrightarrow C_4} E_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{kR_2} E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{kC_2} E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \xrightarrow{kR_2} E_2A \qquad A \xrightarrow{kC_2} AE_2$$

Scaling

- Type 2. A matrix obtained by multiplying a row of I by a nonzero real number. E.g.,

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad k \times R_2, k \neq 0$$

A

$\downarrow kR_2$

$$E_2A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$AE_2 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & ka_{12} & a_{13} \\ a_{21} & ka_{22} & a_{23} \\ a_{31} & ka_{32} & a_{33} \end{bmatrix}.$$

Multiplying on the left by E2 multiplies R2 of A by k and multiplication by E2 on the right multiplies C2 of A by k.

Replacement.

- Type 3. A matrix obtained by adding a nonzero multiple of one row of I to another row, e.g.,

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \qquad R_2 + k \times R_3, k \neq 0$$

$$E_3 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} + ka_{31} & a_{22} + ka_{32} & a_{23} + ka_{33} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$AE_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & ka_{12} + a_{13} \\ a_{21} & a_{22} & ka_{22} + a_{23} \\ a_{31} & a_{32} & ka_{32} + a_{33} \end{bmatrix}.$$

Multiplying on the left by E3 adds k times R3 of A to R2, and multiplication by E3 on the right adds k times C2 of A to C3.

$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 + kR_3} E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix}$

$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{C_3 + kC_2} E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix}$

$A \xrightarrow{R_2 + kR_3} E_3 A$

$A \xrightarrow{C_3 + kC_2} AE_3$

- In general, suppose E is an $n \times n$ elementary matrix. We can think of E as being obtained from I by either a row operation or a column operation.
- If A is an $n \times r$ matrix, **pre-multiplying** A by E (i.e., EA) has the effect of performing that same row operation on A.
- If B is an $m \times n$ matrix, **post-multiplying** B by E (i.e., BE) is equivalent to performing that same column operation on B.

Elementary Matrices

elementary row operation ⇔
left multiply with an elementary matrix

$I_3 \xrightarrow{R_2 \leftrightarrow R_3}$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

R2 ↔ R3

$$\begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{E_1} A = E_1 A$$

5R2

$$\begin{bmatrix} a & b & c \\ 5d & 5e & 5f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = E_2 A$$

$I_3 \xrightarrow{5R_2} E_2$

R2+ 2R1

$$\begin{bmatrix} a & b & c \\ d+2a & e+2b & f+2c \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = E_3 A$$

$I_3 \xrightarrow{R_2+2R_1} E_3$

Eg. $A=\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$ (2×3) $\xrightarrow{R_1\leftrightarrow R_2}$ (E_1) $\begin{bmatrix}4&5&6\\1&2&3\end{bmatrix}$ (E_1A, 2×3) $\xrightarrow{R_1-R_2}$ (E_2)

$\begin{bmatrix}3&3&3\\1&2&3\end{bmatrix}$ $\xrightarrow{2R_2}$ (E_3) $\begin{bmatrix}3&3&3\\2&4&6\end{bmatrix}$

$E_2(E_1A)$ $\qquad$ $E_3(E_2E_1A)$

Write down the elementary matrices for each operation.

$I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix}\xrightarrow{R_1\leftrightarrow R_2} E_1=\begin{bmatrix}0&1\\1&0\end{bmatrix}$

$E_1=\begin{bmatrix}0&1\\1&0\end{bmatrix}\xrightarrow{R_1-R_2} E_2=\begin{bmatrix}-1&1\\1&0\end{bmatrix}$? ✗

$I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix}\xrightarrow{R_1-R_2} E_2=\begin{bmatrix}1&-1\\0&1\end{bmatrix}$

$I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix}\xrightarrow{2R_2} E_3=\begin{bmatrix}1&0\\0&2\end{bmatrix}$

$A \xrightarrow{E_1} \xrightarrow{E_2} \cdots \xrightarrow{E_k} B$

$B = E_k \cdots E_2E_1A$

Theorem

If E is an elementary matrix, then E is nonsingular and E^{-1} is also an elementary matrix of the same type.

Proof

1. If E is an elementary matrix by interchanging the ith and j^{th} rows of I. Then E can be changed back to I by interchanging the same rows again. $EE = I$.
2. If E is an elementary matrix by multiplying the ith row with nonzero scaler α. Then E can be changed back to I by multiplying the same row with $\frac{1}{\alpha}$ (E).
3. If E is an elementary matrix by adding m times the ith row to j^{th} row. Then E can be changed back to I by subtracting m times the ith row from j^{th} row .

1. Theorem

(1): E $n \times n$, nonsingular.

(2) E corresponds to one elementary row/column operation.

Eg. E is of Type I, E^{-1} is of type I.

(3) E^{-1} corresponds to one elementary row/column operation of same type.

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2. $E^{-1}E = I$

$$A = IA = E^{-1}EA = E^{-1}(EA)$$

$$A \xrightarrow{E} EA \xrightarrow{E^{-1}} E^{-1}(EA) = A$$

(1) Type I: Interchange

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow[E]{R_1 \leftrightarrow R_2} EA = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix} \xrightarrow[E^{-1}]{R_1 \leftrightarrow R_2 \ ?} A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 3$

$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} E^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Type I: E: $R_i \leftrightarrow R_j$ $\quad$ E^{-1}: $R_i \leftrightarrow R_j$

$E^{-1} = E$

② Type II: Scaling.

Eg. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{2R_1} \begin{bmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

E $\quad$ EA $\quad$ E^{-1} $\quad$ A

$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{2R_1} E = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ ← diagonal

$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1} E^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}$ ← diagonal

Type II: Scaling

E: αR_i $\quad \alpha \neq 0$ $\quad$ E^{-1}: $\frac{1}{\alpha} R_i$

Type III: Replacement.

Eg. $A=\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix} \xrightarrow{R_2+2R_1} \begin{bmatrix}1&2&3\\6&9&12\end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$

R_2+2R_1: E; EA; R_2-2R_1: E^{-1}

$I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix} \xrightarrow{R_2+2R_1} E=\begin{bmatrix}1&0\\2&1\end{bmatrix}$

$I_2=\begin{bmatrix}1&0\\0&1\end{bmatrix} \xrightarrow{R_2-2R_1} E^{-1}=\begin{bmatrix}1&0\\-2&1\end{bmatrix}$

Type III:

E: $R_j+\alpha R_i$ $\quad$ E^{-1}: $R_j-\alpha R_i$

Summary:

Type I: E: $R_i \leftrightarrow R_j$, E^{-1}: $R_i \leftrightarrow R_j$

Type II: E: αR_i, $\alpha \neq 0$ $\quad$ E^{-1}: $\frac{1}{\alpha} R_i$

Type III: E: $R_j+\alpha R_i$ $\quad$ E^{-1}: $R_j-\alpha R_i$

Theorem

If E is an elementary matrix, then E is nonsingular and E^{-1} is also an elementary matrix of the same type.

Type I: $E = E^{-1}$

Type II $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \ddots & 0 & \vdots \\ \vdots & 0 & \alpha & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}$

$$E^{-1} = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & O & \\ & & 1 & & & & \\ & & & 1/\alpha & & & \\ & & & & 1 & & \\ & O & & & & \ddots & \\ & & & & & & 1 \end{bmatrix} \quad i\text{th row}$$

Type III $E = \begin{bmatrix} 1 & & & & & & \\ \vdots & \ddots & & & & O & \\ 0 & \cdots & 1 & & & & \\ \vdots & & & \ddots & & & \\ 0 & \cdots & m & \cdots & 1 & & \\ \vdots & & & & & \ddots & \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix}$ ith row, jth row

$$E^{-1} = \begin{bmatrix} 1 & & & & & & \\ \vdots & \ddots & & & & O & \\ 0 & \cdots & 1 & & & & \\ \vdots & & & \ddots & & & \\ 0 & \cdots & -m & \cdots & 1 & & \\ \vdots & & & & & \ddots & \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix}$$

Row equivalent

Theorem

A matrix B is row equivalent to A if there exists a finite sequence $E_1, E_2, \ldots, E_k$ of elementary matrices such that

$$B = E_k E_{k-1} \cdots E_1 A$$

Properties:

- If A is row equivalent to B, then B is row equivalent to A.
- If A is row equivalent to B, and B is row equivalent to C, then A is row equivalent to C.

$A = E_k \cdots E_2 E_1 B$

$B = E_m \cdots E_{k+2} E_{k+1} C$

$A = E_k \cdots E_k E_m \cdots E_{k+1} C$

Proof ? (exercise)

① $A \xrightarrow{E_1} E_1A \xrightarrow{E_2} E_2(E_1A) \cdots \xrightarrow{E_k} B$

$B = E_kE_{k-1}\cdots E_2E_1A$

$\neq$

$E_1E_2\cdots E_kA$ ✗

② A is row equivalent to B.

$B \xrightarrow{E_1} E_1B \xrightarrow{E_2} \cdots \xrightarrow{E_k} A$

$B \xleftarrow{E_1^{-1}} \xleftarrow{E_2^{-1}} \cdots \xleftarrow{E_k^{-1}} A$

$A = (E_kE_{k-1}\cdots E_2E_1)B$

Theorem: If $A_1, \cdots A_k$ are nonsingular.

$A_1A_2\cdots A_k$ is nonsingular

$(A_1A_2\cdots A_k)^{-1} = A_k^{-1}\cdots A_2^{-1}A_1^{-1}$ ⌟

$E_kE_{k-1}\cdots E_1$ is nonsingular

$(E_k\cdots E_1)^{-1}A = (\ \not{}^{-1}(\ /\ )B$

$B = E_1^{-1}E_2^{-1}\cdots E_k^{-1}A$

B is row equivalent to A.

Theorem

nonsingular.

If *A* and *B* are row equivalent matrices and *A* is invertible, then *B* is invertible.

$B^{-1} = A^{-1}E_1^{-1}E_2^{-1}\cdots E_k^{-1}$

$B = E_k \cdots E_2 E_1 A$

nonsingular nonsingular

The product of nonsingular matrices is still nonsingular.

Proof

If $A \sim \ldots \sim B$, then $B = E_n \cdots E_2 E_1 A$ for some elementary matrices $E_n \cdots E_2$ and E_1.

So

$$B^{-1} = (E_n \cdots E_2 E_1 A)^{-1}$$
$$= A^{-1}E_1^{-1}E_2^{-1}\cdots E_n^{-1}$$ *since A and E_n are invertible*

Theorem

Equivalent Conditions for Nonsingularity

Let A be a $n \times n$ matrix. The following are equivalent.

(a) A is nonsingular.

(b) $A\mathbf{x}=0$ has only the trivial solution.

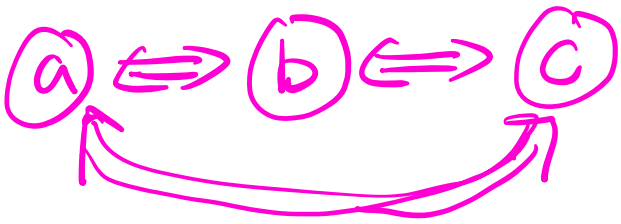
(c) A is row equivalent to I.

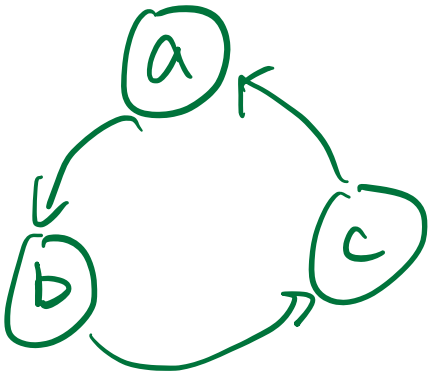
(d) $A\mathbf{x}=\mathbf{b}$ has the unique solution, $\forall \mathbf{b} \in \mathbb{R}^n$

$\mathbf{x}=A^{-1}\mathbf{b}$

Corollary:

The system of n linear equations with n unknowns $A\mathbf{x}=b$ has a unique solution if and only if A is nonsingular.





① a → b

A is nonsingular. A^{-1} exists.

$Ax=0 \rightarrow A^{-1}Ax = A^{-1}0 \rightarrow x = A^{-1}0 = 0$

($Ax=0$: $n \times n$; $A^{-1}A = I$; $A^{-1}0$: unique, gives; 0: unique)

$\rightarrow Ax=0$ has the unique trivial solution.

② b → c

$Ax=0$ has the unique trivial solution.

($Ax=0$: $n \times n$)

⇓

A can be simplified to S.T.F

(consistency theorem)

⇓

The r.r.e.f of S.T.F is I.

⇓

The r.r.e.f of A is I.

||

$\Downarrow$

A is row equivalent to I.

(3) (c) $\rightarrow$ (a)

Since A is row equivalent to I.

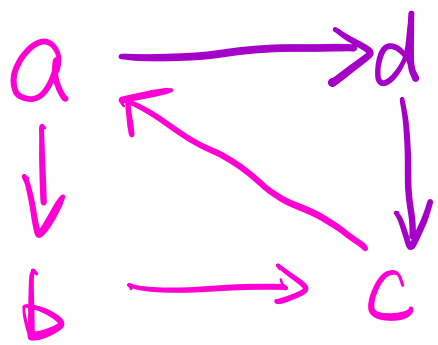
$$A = (E_k E_{k-1} \cdots E_2 E_1) I$$

$$= \underbrace{E_k E_{k-1} \cdots E_2 E_1}$$

nonsingular.

$E_1, E_2, \cdots, E_k$ are all nonsingular.

$$A^{-1} = (E_k E_{k-1} \cdots E_1)^{-1} = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$$



(a) $\rightarrow$ (d) A is nonsingular. A^{-1} exists.

$Ax = b \rightarrow A^{-1}Ax = A^{-1}b \rightarrow x = A^{-1}b$

$A^{-1}A = I$

$x = A^{-1}b$: unique; A^{-1}: unique; b: given

$Ax = b$ has the unique solution.

(d) → (c)

$\underset{n\times n}{A}x = b$ has the unique solution

⇕

A can be simplified to S.T.F

⇓

The r.r.e.f of S.T.F is I

⇓

The r.r.e.f of A is I

⇓

A is row equivalent to I.

Inversely speaking.

A : n×n

If p → q

If not q → not p

Equivalent conditions.

(a) A is singular.

(b) $Ax = 0$ has a nontrivial solution (infinitely

(c) A is not row equivalent to I.

(The r.r.e.f of A is not I).

(d) $Ax = b$ has no solution or infinitely many solutions.

Eg. A: 3×3.

$a_1 = a_2$ ⟶ $1\cdot a_1 - 1\cdot a_2 + 0\cdot a_3 = 0$

Is A nonsingular?

$(1, -1, 0)^T$ is a non-trivial solution to $Ax = 0$ $\Rightarrow$ A is singular.

- If A is nonsingular, then A is row equivalent to I. Therefore there exist elementary matrices $E_1, E_2 \ldots, E_k$ such that

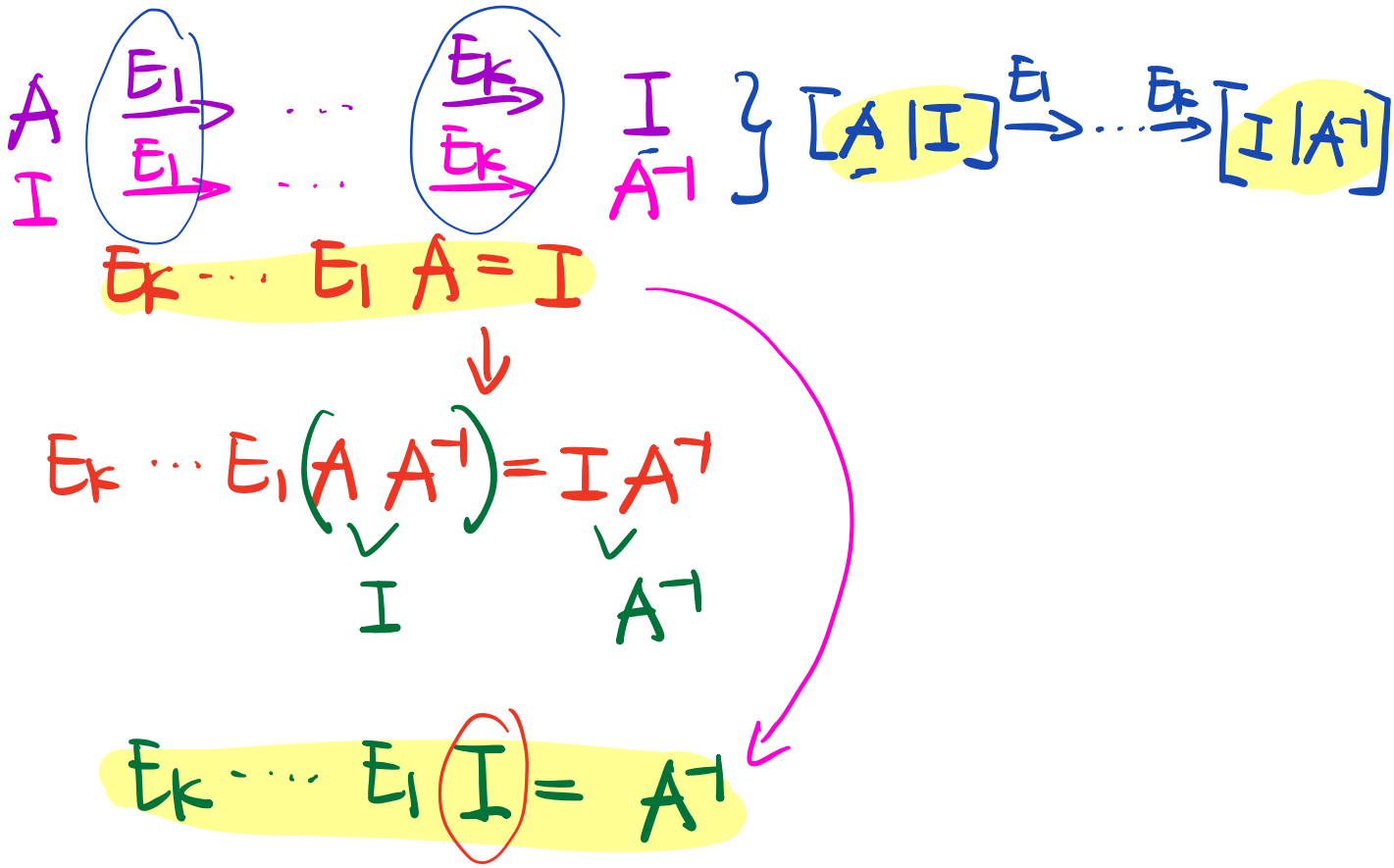
$$E_k E_{k-1} \cdots E_1 A = I$$

- If we multiply on the right by A^{-1}, then

$$E_k E_{k-1} \cdots E_1 I = A^{-1}$$

- Therefore the same row operations that transform A into I, also transform I into A^{-1}. That is to say, the reduced row echelon form of the augmented matrix $[A|I]$ will be $[I\ |A^{-1}]$
- We can use this fact to find the inverse matrix!

A is nonsingular. $\rightarrow$ $A \to I$



Gauss-Jordan Elimination for finding the Inverse of a Matrix

Let A be an $n \times n$ matrix.

1. Adjoin the identity $n \times n$ matrix I_n to A to form the matrix $[A \mid I_n]$.
2. Compute the reduced echelon form of $[A \mid I_n]$.

   If the reduced echelon form is of the type $[I_n \mid B]$, then B is the inverse of A.

   If the reduced echelon form is not of the type $[I_n \mid B]$, i.e. the left-hand side $n \times n$ matrix is not I_n, then A has no inverse.

An $n \times n$ matrix A is invertible if and only if its reduced echelon form is I_n.

Example 2

Determine the inverse of the matrix $A = \begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ 3x3

Solution

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 2 & -3 & -5 & 0 & 1 & 0 \\ -1 & 3 & 5 & 0 & 0 & 1 \end{array}\right]$$

$$\overset{\substack{R2+(-2)R1 \\ R3+R1}}{\sim} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 2 & 3 & 1 & 0 & 1 \end{array}\right]$$

$$\overset{(-1)R2}{\sim} \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 2 & 3 & 1 & 0 & 1 \end{array}\right]$$

$$\overset{\substack{R1+R2 \\ R3+(-2)R2}}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 3 & -1 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & -3 & 2 & 1 \end{array}\right]$$

$$\overset{\substack{R1+R3 \\ R2+(-1)R3}}{\sim} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 5 & -3 & -1 \\ 0 & 0 & 1 & -3 & 2 & 1 \end{array}\right] = A^{-1}$$

Thus, $A^{-1} = \begin{bmatrix} 0 & 1 & 1 \\ 5 & -3 & -1 \\ -3 & 2 & 1 \end{bmatrix}$.

Example 3 (Using Example 2)

Solve the system of equations

$$x_1 - x_2 - 2x_3 = 1$$
$$2x_1 - 3x_2 - 5x_3 = 3$$
$$-x_1 + 3x_2 + 5x_3 = -2$$

$Ax=b$

$\downarrow$

$x=A^{-1}b$

Solution

This system can be written in the following matrix form:

$$\begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = b$$

If the matrix of coefficients is invertible, the unique solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

This inverse has already been found in Example 2. We get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 5 & -3 & -1 \\ -3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

The unique solution is $x_1 = 1, x_2 = -2, x_3 = 1.$

Example 4

Determine the inverse of the following matrix, if it exist.

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 2 & 7 \\ 2 & -1 & 4 \end{bmatrix}$$

Solution

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 1 & 5 & 1 & 0 & 0 \\ 1 & 2 & 7 & 0 & 1 & 0 \\ 2 & -1 & 4 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \begin{matrix} \\ R2+(-1)R1 \\ R3+(-2)R1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 1 & 5 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & -3 & -6 & -2 & 0 & 1 \end{array}\right]$$

$$\sim \begin{matrix} \\ R1+(-1)R2 \\ R3+3R2 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & -5 & 3 & 1 \end{array}\right]$$

A is not row equivalent to I

⇕

A is singular.

There is no need to proceed further.

The reduced echelon form cannot have a one in the (3, 3) location.

The reduced echelon form cannot be of the form $[I_n : B]$.

Thus A^{-1} does not exist.

A: nxn square

Triangular matrices

{ upper
  lower.

- A square $n \times n$ matrix $A = (a_{ij})$ is *upper triangular* if $a_{ij} = 0$ for $i > j$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & a_{nn} \end{bmatrix}$$

(2)

- The products or sums of two upper triangular matrices are still upper triangular. (1)
- Type III elementary matrices corresponding to "upper row – k*lower row" are upper triangular.

$I_3 \xrightarrow{R_1 - 2R_2} E = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Eg. $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Is A upper triangular? ✓

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ Is A upper ——— ?

A, B are both $n \times n$ upper triangular matrices.

① $C = A + B$, C is upper triangular.

Objective: $c_{ij} = 0$ when $i > j$.

When $i > j$ $\quad c_{ij} = a_{ij} + b_{ij}$

Since A, B are both upper triangular.

$a_{ij} = 0$, $i > j$ $\quad b_{ij} = 0$ $i > j$

$\Rightarrow c_{ij} = a_{ij} + b_{ij} = 0 \quad i > j$

$\Rightarrow$ C is upper triangular.

② $C = AB$ is still upper triangular.

$[\ c_{ij} = 0 \quad i > j\ ]$ $\qquad i > j \Rightarrow i-1 \geq j$

$c_{ij} = \vec{a}_i b_j =$

$$[\underbrace{0\ 0 \cdots 0}_{i-1}\ a_{ii}\ a_{i,i+1}, \cdots a_{in}] \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{jj} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$= 0 b_{1j} + 0 b_{2j} + \cdots + 0 b_{i-1,i-1}$

$+ a_{ii} \cdot 0 + \cdots + a_{in} \cdot 0 = 0 \qquad i > j$

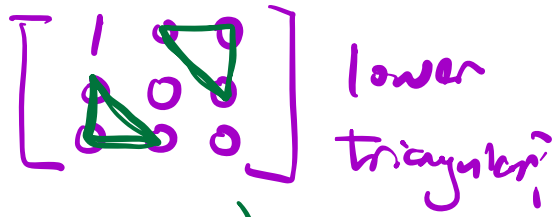
$\Rightarrow C$ is upper triangular.

$\Rightarrow$ General case. $A_1, \cdots, A_k$ upper triangular.

$\Rightarrow A_1 + A_2 + \cdots + A_k$ is strictly upper —

$\Rightarrow A_1 A_2 \cdots A_k$ is strictly upper —

Triangular matrices



nxn A

- It is *lower triangular* if $a_{ij} = 0$ for $i < j$.

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & 0 \\ a_{n1} & a_{n2} & a_{n3} & \cdots & \cdots & a_{nn} \end{bmatrix}$$

both lower & upper.

diagonal

- The products or sums of two lower triangular matrices are still lower triangular.
- Type III elementary matrices corresponding to "lower row – k*upper row" are lower triangular.

$I_3 \xrightarrow{R_2 - 2R_1} E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Diagonal matrices

- *A diagonal matrix* is both upper and lower triangular, that is to say $a_{ij} = 0$ for $i \neq j$.

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & \cdots & 0 \\ 0 & 0 & a_{33} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & 0 \\ 0 & 0 & 0 & \cdots & \cdots & a_{nn} \end{bmatrix}$$

- A diagonal matrix can be written as a product of elementary matrices of type II. For example:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$



LU decomposition (optional)

- If an $n \times n$ matrix A can be reduced to strict upper triangular form U, using only row operation III (lower row minus multiple of upper rows), then $A = LU$, where L is lower triangular and U is upper triangular.
- Proof: If $E_k \ldots E_2 E_1 A = U$ where E_i are lower triangular elementary matrices, and U is upper triangular, then

$$A = E_1^{-1} E_2^{-1} \ldots E_k^{-1} U = LU$$

  Where $L = E_1^{-1} E_2^{-1} \ldots E_k^{-1}$ is lower triangular.
- *Example: see book exercise 6*
- LU decomposition of large matrices is very important in engineering applications.

$A = LU$

L: lower triangular

U: upper triangular

$A \to U$: Type III (adding a multiple of upper rows into the lower rows)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow[R_3 - 7R_1]{R_2 - 4R_1}$$

$U = (E_k E_{k-1} \cdots E_2 E_1) A$

lower

$(E_k E_{k-1} \cdots E_2 E_1)$: a lower triangular matrix

$(E_k \cdots E_1)^{-1} U = A$

$A = (E_1^{-1} \cdots E_k^{-1}) U = LU$

$E_1^{-1} \cdots E_k^{-1}$: lower triangular

Eg. $E = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

$E^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

let $L = E_1^{-1} \cdots E_k^{-1}$ lower triangular

Summary

- 3 types of elementary matrices correspond to 3 types of row/column operations.
  - Multiplying elementary matrices on the left corresponds to row operations,
  - Multiplying elementary matrices on the right corresponds to column operations.
- Matrix A is nonsingular if and only if A is row equivalent to I_n.
  - To calculate the inverse matrix of a nonsingular A, write down $[A|I_n]$ and then row reduced it to $[I_n|A^{-1}]$.
- There are special matrices called upper/lower triangular matrices and diagonal matrices. Many matrices can be written as a lower triangular matrix times an upper triangular matrix.